



Assessed Coursework

Course Name	Database & Data Analytics 2025			
Coursework Number	1/1			
Deadline Time:		16h30	Date:	27/11/2025
% Contribution to final course mark	20%			
Solo or Group	✓ Solo	✓	Group	
Anticipated Hours	Average 20 hours			
Submission Instructions	Submission of only <u>one</u> PDF document; please read in the description.			
Please Note: This Coursework cannot be Re-Assessed				

Code of Assessment Rules for Coursework Submission

Deadlines for the submission of coursework which is to be formally assessed will be published in course documentation, and work which is submitted later than the deadline will be subject to penalty as set out below.

The primary grade and secondary band awarded for coursework which is submitted after the published deadline will be calculated as follows:

- (i) in respect of work submitted not more than five working days after the deadline
 - a. the work will be assessed in the usual way;
 - b. the primary grade and secondary band so determined will then be reduced by two secondary bands for each working day (or part of a working day) the work was submitted late.
- (ii) work submitted more than five working days after the deadline will be awarded Grade H. Penalties for late submission of coursework will not be imposed if good cause is established for the late submission. You should submit documents supporting good cause via MyCampus.

Penalty for non-adherence to Submission Instructions is: 2 bands

Database in Theory & Practice 2025

Task 1: Basic Relational Schema Modelling [Marks: 40]

We are about to store information of lecturers working at a university. For each lecturer, we need to store their Social Security Number (SSN), which is unique, their SURNAME, their home ADDRESS, the LANGUAGE(s) they can speak (it can be more than one language), the courses they can teach (COURSES), e.g., programming, databases, data science, their SALARY per annum, and their DEPARTMENT. We initially store this information as shown in Table 1 (this is an instance/example with 3 lecturers).

For instance: McDonald is the lecturer with SSN = 1 whose home address is 35 St. Vincent. The courses McDonald can teach are 'Programming' and 'Databases', he can speak three languages: 'Dutch', 'English', and 'Mandarin', his salary is 51,000 GBP, and works in the Computing department.

Table 1. Instance with lecturers.

SSN	SURNAME	ADDRESS	LANGUAGE	TEACHES	SALARY	DEPARTMENT
1	MCDONALD	35 St. Vincent	{Dutch, English, Mandarin}	{Programming, Databases}	51,000	Computing
2	BROOKS	4 Hyndland Rd	{English, French}	{Programming, Databases}	55,000	Engineering
3	WONG	1 University Ln	{Mandarin, English}	Databases	50,000	Computing

Our aim is to store lecturers' information to a relational database, thus, we initially store this information in a relation which is (at least) in 1NF. We then obtain the relation LECTURER in Table 2, where we decide to have as (composite) Primary Key, the attributes: PK = {SSN, LANGUAGE, TEACHES} (these prime attributes are also underlined in Table 2). There is no dependency between LANGUAGE and TEACHES attributes.

Table 2: LECTURER Relation

<u>SSN</u>	SURNAME	ADDRESS	<u>LANGUAGE</u>	<u>TEACHES</u>	SALARY	DEPARTMENT
1	MCDONALD	35 St. Vincent	Dutch	Programming	51,000	Computing
1	MCDONALD	35 St. Vincent	English	Programming	51,000	Computing
1	MCDONALD	35 St. Vincent	Mandarin	Programming	51,000	Computing
1	MCDONALD	35 St. Vincent	Dutch	Databases	51,000	Computing
1	MCDONALD	35 St. Vincent	English	Databases	51,000	Computing
1	MCDONALD	35 St. Vincent	Mandarin	Databases	51,000	Computing
2	BROOKS	4 Hyndland Rd	English	Programming	55,000	Engineering
2	BROOKS	4 Hyndland Rd	French	Programming	55,000	Engineering

2	BROOKS	4 Hyndland Rd	English	Databases	55,000	Engineering
2	BROOKS	4 Hyndland Rd	French	Databases	55,000	Engineering
3	WONG	1 University Ln	Mandarin	Databases	50,000	Computing
3	WONG	1 University Ln	English	Databases	50,000	Computing

Task 1.1: The relation LECTURER in Table 2 is expected to be inefficient w.r.t. insertion, deletion, and update. Assume that we are about to modify the LECTURER relation with modifications in the following order:

- **M1:** We need to update McDonald's record by adding a new language: 'Japanese'.
- **M2:** We need to update the home address of Brooks with a new one.
- **M3:** We need to add the new course for Brooks, which is: 'Data Science'.
- **M4:** We insert 100 new lecturers with the following information: All these new lecturers teach the same three courses: 'Databases', 'Programming', and 'Data Science'. They all also speak two languages: 'English' and 'Dutch'. (the home addresses and salaries can be any values)
- **M5:** We need to delete the 'Programming' course from McDonald.

Provide the number of tuples we need to add, or update, or delete for each of the above-mentioned modifications (M1 – M5) and provide a short explanation about the anomaly of each modification. **[5 Marks] 1 mark for each modification.**

Task 1.2: Using the pgAdmin4 application, create the relation LECTURER and insert the lecturer tuples as shown in Table 2, i.e., **before** the modifications from Task 1.1. Then, provide SQL statements for the following queries over the LECTURER:

- **SQL1:** Show the number of courses taught per lecturer.
- **SQL2:** How many languages does Brooks (SSN = 2) speak?
- **SQL3:** How many lecturers speak Mandarin?
- **SQL4:** How many lecturers teach more than one course?
- **SQL5:** From those lecturers teaching more than one course, how many earn more than 50,000?
- **SQL6:** From those lecturers who can speak more than 2 languages, who teach the 'Databases' course (which are their SSNs and surnames)?
- **SQL 7:** For each department with more than 2 lecturers, how many of them earn more than 51,000 and speak more than 2 languages?

Which of the above-mentioned SQL queries was the most challenging? Explain briefly your answer. **[35 Marks] SQL1: 1 mark; SQL2: 1 mark; SQL3: 1 mark; SQL4: 2 marks; SQL5: 5 marks; SQL6: 10 marks; SQL7: 15 marks.**

Task 2: Theory of Multi-valued Dependencies & 4NF [Marks: 60]

As you might have witnessed from Task 1, the LECTURER relation causes certain modification anomalies and challenges.

Task 2.1: Identify the maximum NF of the LECTURER (Table 2) and explain your reasoning. **[1 Mark]**

Task 2.2: From the LECTURER relation, we can observe that there are certain *many-to-many* relationships between attributes. For instance, lecturers (SSN) teaching courses, and lecturers (SSN) with languages.

That is, an lecturer can teach more than one course, while a course can be taught by more than one lecturer. Similarly, an lecturer can speak many languages, while a language can be spoken by more than one lecturer (obviously). These two many-to-many relationships (lecturer-course; lecturer-language) are *independent*, in the sense that there is no direct connection between courses taught and languages. Hence, these specific dependencies among the attributes: lecturer, course, and language (all belonging in the primary key) cause modification anomalies and challenges in Task 1. These dependencies are called *Multi-Valued Dependencies* (MVDs) in the database literature. In this task, you are asked to search information about the MVDs, as generalizations of the well-known Functional Dependencies, and the associated *Fourth Normal Form* (4NF) of a relation. Information related to 4NF and MVDs can be found in:

- [1] https://en.wikipedia.org/wiki/Fourth_normal_form (Forth Normal Form; Wikipedia)
- [2] https://en.wikipedia.org/wiki/Multivalued_dependency (Multi-valued Dependency; Wikipedia)
- [3] <http://tinyurl.com/ycz3rgun> ('Database Systems: The Complete Book', H. Garcia-Molina, J.D. Ullman and J. Widom, 2nd, Pearson, 2014). Read (online) pages from the eBook (GUID log-in is required): 'Design Theory for Relational Databases', Chapter 6: Multivalued Dependencies (only pages: 101—108).

Provide a 500-word document (approximately 2/3 of a page) explaining the MVDs and the 4NF. Provide also an example (not the one described here).

[15 Marks]

Task 2.3: Identify the MVDs in the LECTURER relation in Table 2 and split this relation into relations being in 4NF. Explain your process including the definition of PKs and FKs in the derived relations. **[4 Marks]**

Task 2.4: Repeat the Task 1.1 (i.e., how many tuples are needed to be inserted/modified for the M1— M5 modifications) over your new relational schema being in 4NF (from Task 2.3). **[5 Marks] 1 mark for each modification: M1 – M5.**

Task 2.5: Provide the SQL statements of the Task 1.2 (for all the queries: SQL1 – SQL7) over your new relational schema being in 4NF (from Task 2.3). For each SQL statement, discuss any modification you made with the corresponding SQL statement in Task 1.2.

[35 Marks] SQL1: 1 mark; SQL2: 1 mark; SQL3: 1 mark; SQL4: 2 marks; SQL5: 5 marks; SQL6: 10 marks; SQL7: 15 marks.

Submission

Answer to *all* the tasks and submit only **one PDF file** (including your GUID).

Note 1: Add the corresponding numbers of the tasks answered. **Note**

2: Good luck!